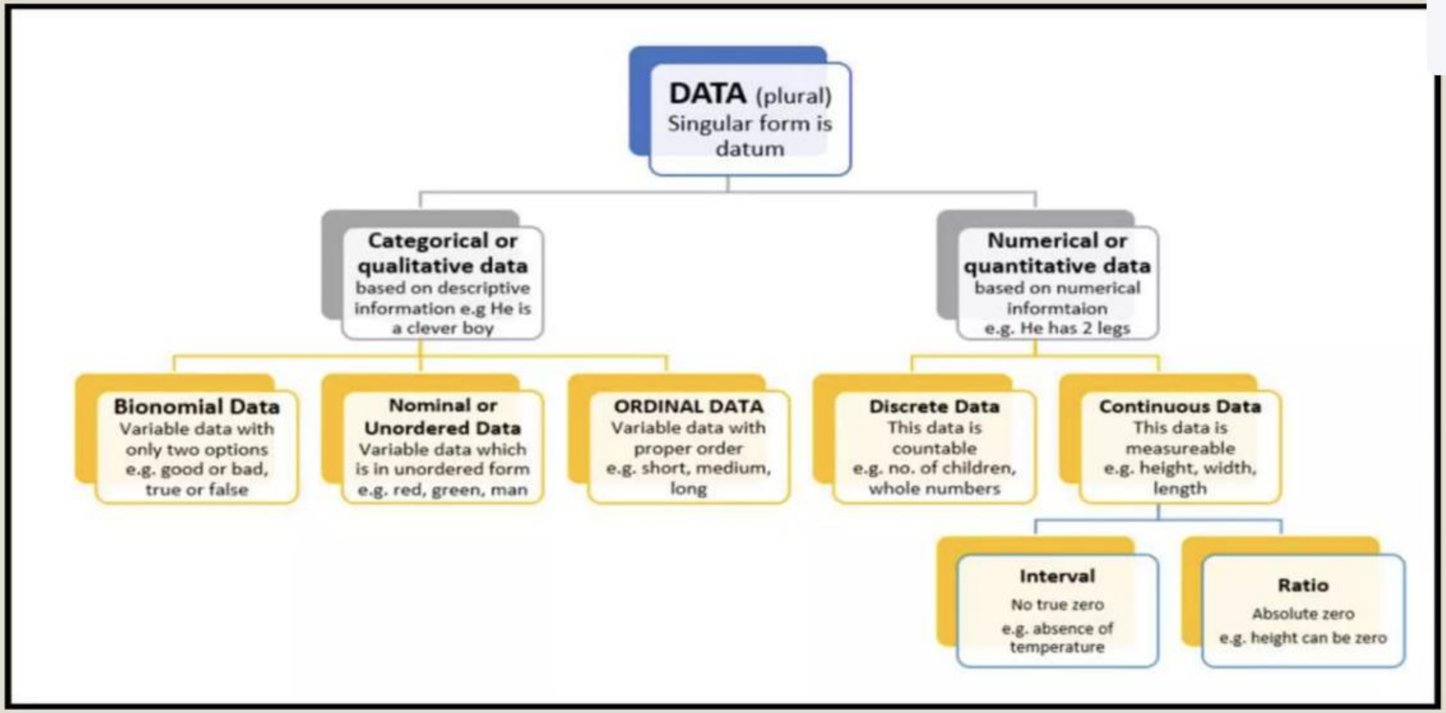
Unit 3: Data

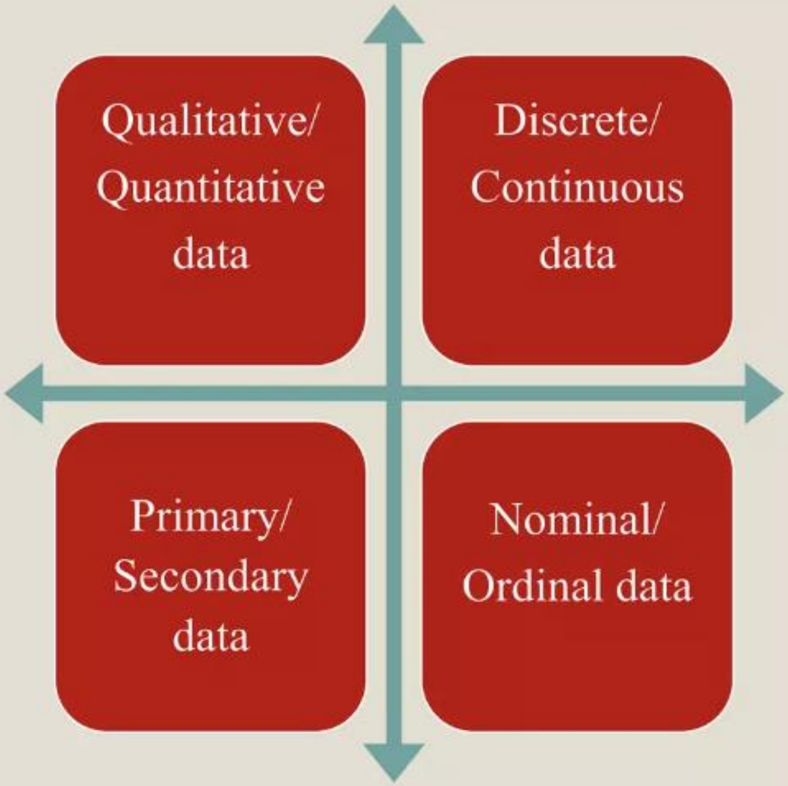
INTRODUCTION

- Data are individual units of information. A data describes a single quality or quantity of some object or phenomenon. In analytical processes, data are represented by variables.
- Data presentation is a method by which people organize, summarize and communicate information using a variety of tools such as text , tables , graphs and diagrams

CLASSIFICATION



TYPES OF DATA



Qualitative data:

Also called as enumeration data .Represents a particular quality or attribute. There is no notion of magnitude or size of the characteristic, as they can't be measured. Expressed as numbers without unit of measurements .

Eg: Religion, Sex, Blood group etc.

Quantitative data:

Also called as measurement data. These data have a magnitude. Can be expressed as number with or without unit of measurement.

Eg: Height in cm, Hb in gm%, BP inmm of Hg, Weight in kg.

DISCRETE DATA:

Here we always get a whole number.

Eg. Number of beds in hospital, Malaria cases

CONTINUOUS DATA :

It can take any value possible to measure or possibility of getting fractions.

Eg. Hb level, Ht, Wt.

PRIMARY DATA :

Obtained directly from an individual , it gives precise information .

SECONDARY DATA :

Obtained from outside source

Eg: Data obtained from hospital records, Census

NOMINAL DATA:

The information or data fits into one of the categories, but the categories cannot be ordered one above another .

E.g.. Colour of eyes, Race, Sex

ORDINAL DATA:

here the categories can be ordered, but the space or class interval between two categories may not be the same.

E.g.. We could have categories for prognosis such as good, fair, poor, hopeless, or stages of periodontitis as mild, moderate, or severe

UNPAIRED (INDEPENDENT OR UNMATCHED) DATA

Where data are obtained from two groups that are unrelated to each other. Measurements are taken on two separate groups of individuals.

E.g. males vs. females, age groups, and parallel designs.

PAIRED OR MATCHED DATA

where the measurements are taken on the same individual or matched groups as in a split mouth or same group before and after or cross over designs.

USES OF DATA PRESENTATION METHODS

- Easy and better understanding of the subject
- Provides first hand information about data
- Helpful in future analysis
- Easy for making comparisons

CRITERIA FOR SELECTING A DATA PRESENTAION METHOD

- Size of study
- Scope of study
- Program participation
- Worker cooperation
- Intrusion into the lives of research participants
- Resources
- Time
- Previous research findings

Representation and Input

- Data representation and input are fundamental to processing and storage in computers.
- Key stages include:
- 1. Collection
- 2. Preparation
- 3. Verification
- 4. Input Methods

Stages of Data Handling

1. Collection: Gathering raw data from various sources.
2. Preparation: Cleaning and organizing data for input.
3. Verification: Ensuring data correctness.
4. Input Methods: Manual entry, scanning, and automated processes.

Input Devices and Media

Input Devices:
- Keyboard
- Mouse
- Scanner
- Microphone

Input Media:
- Optical Discs
- Magnetic Disks
- Flash Drives

What are Peripherals?

- **Definition:** Devices that are not part of the central computer but enhance its functionality.
- **Examples:** Printers, keyboards, scanners, external hard drives.

Online Peripherals

- **Definition:** Devices directly connected to and controlled by the computer in real-time.
- **Features:**
  - Operates in sync with the computer system.
  - Provides immediate access and functionality.
- **Examples:**
  - **Keyboard:** Real-time data input for text and commands.
  - **Monitor** Displays output instantly as the computer processes data.
  - **Webcam:** Streams video directly to the computer.

Offline Peripherals

- **Definition:** Devices that operate independently or store data for later use, not requiring constant connection to the computer.
- **Features:**
  - Perform tasks offline, such as data collection or printing.
  - Can be connected to transfer or retrieve data when needed.
- **Examples:**
  - **Printer:** Collects data from the computer and prints it later.
  - **External Hard Drives:** Stores data offline and connects only when needed.
  - **Scanners:** Operates independently to capture images or documents.

Online and Offline Peripherals

Online Peripherals:

- Directly connected and active during operation.
- Examples: Printers, External Hard Drives.

Offline Peripherals:

- Not connected in real-time.
- Examples: USB Drives, Backup Tapes.

Comparison Between Online and Offline Peripherals

Feature	Online Peripherals	Offline Peripherals
Connectivity	Always connected	Connected as needed
Operation	Real-time interaction	Independent operation
Examples	Keyboard, webcam	Printer, scanner
Advantage	Immediate functionality	Portability and flexibility
Disadvantage	Requires constant link	Delayed interaction

What is Verification?

- Verification checks **if the data has been accurately entered and transferred** from one stage or system to another. It ensures the data is consistent and free of errors during data entry or migration.

What is Validation?

- Validation ensures **the data meets predefined rules and constraints** before being processed or stored. It checks for logical consistency and adherence to standards.

Verification and Validation

Verification:

- Ensures data is copied correctly.
- Examples: Double Entry, Proofreading.

Validation:

- Ensures data is logical and complete.
- Examples: Range Check, Format Check.

Differences Between Verification and Validation:

Aspect	Verification	Validation
Purpose	Checks data entry accuracy.	Ensures data meets requirements.
Timing	During data entry or transfer.	During or before data processing.
Methodology	Comparison or review.	Logical rules and checks.

Importance of Verification and Validation:

1. Prevents errors in data processing.
2. Enhances data reliability and accuracy.
3. Reduces operational risks and decision-making errors.
4. Ensures compliance with regulations and standards.

Digital Symbols and Number Bases

Digital Symbols:

- Represent data in binary form (0s and 1s).

Number Bases:

- Binary (Base 2): Used in computers.
- Decimal (Base 10): Used in daily life.
- Hexadecimal (Base 16): Used in coding and memory addressing.

Coding Schemes for Character Storage

- Common Coding Schemes:
- ASCII: Represents English characters (7-bit).
- Unicode: Supports multiple languages (16-bit).
- EBCDIC: Used in mainframes.

- Examples:
- ASCII: A = 65, B = 66.
- Unicode: Supports Hindi, Chinese, and more.

Importance of Verification and Validation

Verification:

- Reduces data entry errors.
- Ensures accurate data transfer.

Validation:

- Prevents logical errors.
- Ensures compliance with rules (e.g., date formats).

Summary

- Data representation involves collection, preparation, verification, and input.
- Input devices and media are critical for data entry.
- Verification and validation ensure data accuracy and integrity.
- Digital symbols and coding schemes standardize data storage.